Research Article | Vol. 33, No. 12 / 16 Jun 2025 / Optics Express 26342
Optics EXPRESS

# Volatile organic compound photoionization utilizing tunable quasi-monochromatic vacuum ultraviolet down to 120 nm obtained by third harmonic generation from a suspended dielectric nanomembrane

RAFAEL VILARINHO,[1,†] HEDIEH MAHMOODNIA,[1,†] CHRISTIAN MAIBOHM,[1] DIOGO POETA,[1,2] JOSÉ FERNANDES,[1] CARLOS CALAZA,[1] ANA I. GÓMEZ-VARELA,[3] MARTÍN LÓPEZ-GARCIA,[4] ADELAIDE MIRANDA,[1,5] AND PIETER A. A. DE BEULE[1,6]

[1]*International Iberian Nanotechnology Laboratory, Avenida Mestre José Veiga, s/n, 4715-330 Braga, Portugal*
[2]*Biomolecular Photonics Research Group, Faculty of Physics, Bielefeld University, Bielefeld, Germany*
[3]*Photonics4Life Research Group, Departamento de Física Aplicada, Facultade de Física and Facultade de Óptica e Optometría, Instituto de Materiais (iMATUS), Universidade de Santiago de Compostela, Campus Vida, Santiago de Compostela E15782, Spain*
[4]*Instituto de Óptica, IO-CSIC, Serrano 121, Madrid, 28006, Spain*
[5]*adelaide.miranda@inl.int*
[6]*pieter-de.beule@inl.int*
[†]These authors contributed equally to this work.

**Abstract:** We present a table-top light source of continuously tunable quasi-monochromatic vacuum ultraviolet (VUV) coupled to a custom-built photoionization detector and demonstrate its capacity to induce photoionization in trace levels of volatile organic compounds at ambient saturated vapor pressure. A high-power pulsed monochromatic laser powers an optical parametric amplifier that subsequently feeds into two materials inducing consecutively second harmonic generation (SHG) and third harmonic generation (THG). While the SHG is obtained in a bulk BBO crystal, the THG originates from suspended $SiO_2$ nanomembranes. We demonstrate the photoionization of ethanolamine and eucalyptol at THG pump wavelengths of 390 nm and 360 nm respectively, providing evidence of VUV light generation at 130 nm and down to 120 nm, going beyond the state-of-the-art of demonstrated VUV generation in dielectric nanomembranes. This proof-of-concept system highlights the transformative potential of dielectric nanomembranes in facilitating compact, efficient and tunable VUV light sources for spectroscopy and sensor technology without the necessity of vacuum technology.

© 2025 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

## 1. Introduction

The vacuum ultraviolet (VUV) wavelength domain refers to optical wavelengths below 200 nm, corresponding to photon energies higher than 6.2 eV [1–3]. Aside from research into photoionization and photoelectron spectroscopy [4,5], VUV light finds practical applications in gas sensing [6] and lithography [7,8]. Unfortunately, practical applications of VUV are severely limited by the availability of suitable VUV light sources. Discharge lamps with noble gas fills [9] and excimer/exciplex lamps [10] are the only non-coherent small-scale light sources available. They typically provide continuous wave light output of limited intensity at a single wavelength determined by the noble gas fill or excited dimer (excimer) or complex (exciplex). Coherent VUV light is available from excimer lasers [11] and laser systems incorporating optically non-linear

#564577 https://doi.org/10.1364/OE.564577
Journal © 2025 Received 8 Apr 2025; revised 29 May 2025; accepted 31 May 2025; published 11 Jun 2025

Optics EXPRESS

processes, where-by only the latter can provide broadband VUV light. Finally, free-electron lasers (FELs) have also been demonstrated as an extremely bright source of broadband and ultrafast VUV [12]. Even though FELs are extremely unique and used in large facilities for research purposes, a table-top design for a FEL as VUV source has been contemplated [13,14]. Among the many applications of VUV light, gas sensing stands out as a critical area, particularly in the detection of volatile organic compounds (VOCs) [15]. Photoionization is a widely adopted technique for VOC detection, leveraging the ability of VUV photons to ionize molecules while minimizing molecular fragmentation. The core technology enabling this process, the photoionization detector (PID), integrates a VUV light source, an ionization chamber, and low-current detection electronics [6]. These systems are portable, energy-efficient, and widely used in hazardous material response scenarios. PIDs generate electron-ion pairs upon the absorption of high-energy photons, which must exceed the molecule's first ionization potential (IP), a range spanning from 3.894 eV (Cs) [16] to 24.587 eV (He) [17]. VOCs are predominantly characterized by molecule specific IPs between 8 eV to 11 eV [18], corresponding to 155.0 nm to 112.7 nm respectively. Hence, gas detection through photoionization requires light sources emitting in the latter wavelength range.

High-energy photons are typically generated using low-pressure discharge lamps that emit light between 105 nm (11.8 eV) and 150 nm (8.4 eV), depending on the gas fill. However, the monochromatic nature of traditional VUV light sources in PIDs limits their molecular selectivity, making it impossible to identify specific contaminants, which is a critical limitation for real-time environmental monitoring and industrial applications. Table-top laser systems based on non-linear optical media provide a promising solution, particularly those employing solid-state frequency conversion [19]. However, these systems are restricted to wavelengths above 160 nm due to the transmission cut-off of the bulk materials used for frequency conversion. Recently, nanoscale dielectric membranes, such as those made from $SiO_2$ and $Si_3N_4$, have emerged as a medium for generating shorter VUV wavelengths. Konishi et al. [20] demonstrated that these suspended dielectric nanomembranes can facilitate third harmonic generation (THG) down to 146 nm without the phase-matching constraints that traditionally limit non-linear optical processes in bulk materials. Furthermore, the high damage threshold of these membranes allows for increased excitation power, enhancing overall VUV generation efficiency. Their commercial availability, ease of handling, and compact nature make these nanomembranes an attractive alternative for next-generation VUV light sources.

We present a table-top experimental set-up designed to explore the capacity of THG in the VUV, generated by suspended $SiO_2$ nanomembranes, to perform photoionization of VOCs using an optoelectronic device (Fig. 1). Employing a pulsed femtosecond laser operating at 60 kHz repetition rate with pulses of 230 fs duration and energies of ~4 μJ in the 340 nm to 440 nm range, we generate VUV through THG by illuminating 600 nm thick suspended $SiO_2$ nanomembranes. We outline a device design that enables the unequivocal demonstration of photoionization of two VOCs, ethanolamine and eucalyptol, with THG pump wavelengths of 390 nm and 360 nm respectively, implying the generation of VUV at 130 nm and 120 nm. Hereby, we demonstrate a proof-of-concept system for real-time, selective VOC gas sensing of multiple compounds in complex environments enabled through VUV wavelength tuning.

Optics EXPRESS

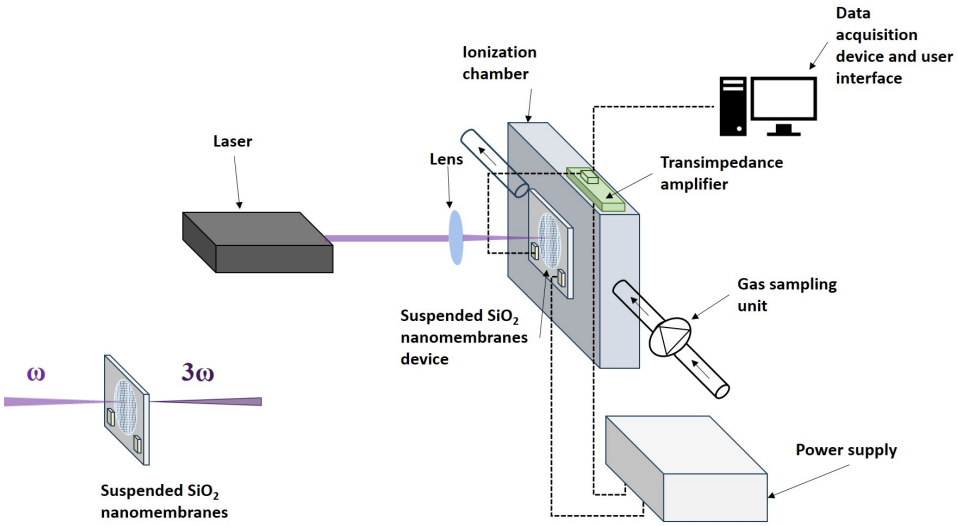



**Fig. 1.** Schematic design of the experimental set-up. A femtosecond laser induces the formation of THG on suspended $SiO_2$ nanomembranes integrated in a device that comprises aluminium metal plates creating an electric field to amplify the photoionization process. Additionally, contact pads are incorporated into the device, enabling the detection of the photoionization current. A saturated vapor pressure of volatile organic compounds in the gas sampling units is passed into the ionization chamber. Dashed lines correspond to electrical connections and arrows to the gas flow.

## 2. Methods

### 2.1. Suspended nanomembranes for THG

We opt for $SiO_2$ as the material of choice for THG generating nanomembranes due to their high reported damage threshold to laser induced damage [20]. The thickness of dielectric nanomembranes is critical for efficiency of the THG process. To correlate the intensity of the THG ($I^{3\omega}$) with the thickness of the suspended $SiO_2$ nanomembranes we carried out theoretical simulations by two approaches: via an analytical expression and via a finite-difference time-domain (FDTD) method.

### 2.2. Device design

A plurality of circular $SiO_2$ nanomembranes with a diameter of 1 mm and 600 nm thickness are arranged in various arrays in a single device based on a silicon wafer support. This allows us to overcome accidental breaking of a nanomembrane by displacing the device with respect to the fixed laser beam path. Furthermore, the device integrates two AlSiCu metal plate electrodes with connected electrical wire contact pads to enable the capture of small photoionization currents. The final device is shown in section 1 of Supplement 1 while Fig. 2 depicts the scheme of the final device and process flow.

### 2.3. Device fabrication

The fabrication process used for the suspended $SiO_2$ nanomembranes is depicted in Fig. 2. It makes use of a double side polished 8 inch silicon (Si) wafer with 725 µm thickness as substrate for the device implementation. The process starts with the deposition of a 3 µm thick $SiO_2$ layer on the backside of the wafer by using plasma enhanced chemical vapor deposition (PECVD) (step 1), intended to act as hard mask for the silicon etch process. The first lithography is then

Optics EXPRESS

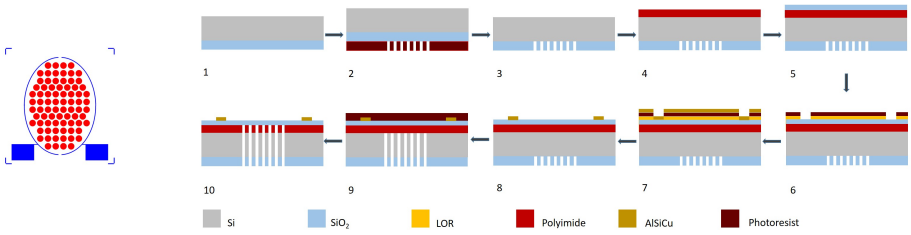



**Fig. 2.** Schematic figure of the final device and process flow: (1) deposition of 3 µm PECVD $SiO_2$; (2) deposition of photoresist DWL lithography; (3) etching $SiO_2$; (4) deposition of polyimide; (5) PECVD deposition of 600 nm $SiO_2$; (6) deposition of LOR and photoresist and DWL lithography; (7) deposition of 250 nm AlSiCu alloy; (8) lift-off process; (9) removing polyimide layer and protection resist layer to achieve suspended membrane.

performed on the backside of the wafer (step 2) to outline the nanomembranes pattern. The thick $SiO_2$ hard mask layer is etched using an Inductively Coupled Plasma−Reactive Ion Etching (ICP−RIE) process with fluorine chemistry (step 3). After resist strip by ashing, the process flow continues on the frontside of the wafer (steps 4−7). First, a sacrificial polyimide (PI) layer is spin coated and then cured at 250 °C for 14 h to achieve a 2 µm thick film. Then a 600 nm thick $SiO_2$ layer is deposited by PECVD to define the thickness of the nanomembranes (step 5). A bilayer of lift-off resist (0.5 µm LOR−5B) and UV photoresist (1.035 µm AZ1505) is spin coated on the frontside of the wafer for the second lithography (step 6), used to outline the AlSiCu metal pads. A 250 nm thick AlSiCu metal layer is deposited by sputtering (step 7) and the metal pads are patterned with a lift-off process. A layer of photoresist is then coated as a protection for the frontside, and the process moves to the backside of the wafer, where the silicon is etched using a Deep Reactive Ion Etching (RIE) process until reaching the polyimide film acting as an etch stop layer (step 9). Finally, the free standing 600 nm thick $SiO_2$ suspended nanomembranes are released by using an isotropic plasma ashing process to remove simultaneously the polyimide sacrificial layer on the backside and the photoresist protective layer on the frontside of the membranes (step 10).

### 2.4. *Integration*

Figure 1 depicts a schematic layout showing the integration of the different components of the experimental set-up. A high-power ultrafast Ytterbium laser (Carbide, Light Conversion, Lithuania) generates femtosecond duration pulses at kilohertz rates for pumping an optical parametric amplifier (OPA) (Orpheus, Light Conversion, Lithuania). In a subsequent step, the tunable quasi-monochromatic wavelength output of OPA is frequency doubled by a $\beta$ phase barium borate (BBO) crystal (Lyra, Light Conversion, Lithuania) via second harmonic generation (SHG). This laser configuration generates femtosecond light pulses over the 315 nm to 540 nm wavelength range with an associated energy of ~4 µJ. These SHG femtosecond light pulses are then focused by a lens (Thorlabs, LA1509−A, AR coating 300 nm to 480 nm), into the suspended $SiO_2$ nanomembranes. The ionization chamber, designed and built in-house, consists of a block of anodized aluminum with dimensions of $80 \times 80 \times 30$ mm (height x length x width) with an airtight volume with gas inlet and outlet, and a chamber volume of 2427.3 $mm^3$. The laser beam is focused to a spot < 100 µm by the $CaF_2$ focusing lens on one nanomembrane of the array at a time. In addition, the positive side of the power supply (Stanford Research Instruments Inc., Model PS 350/5000 V −25 W) is connected to the ionization chamber which is further connected to the transimpedance amplifier circuit board. The positive side of the transimpedance amplifier circuit, a Printed Circuit Board (PCB), is connected to the negative side of the power supply. The transimpedance precision amplifier consists of an amplifier with

Optics EXPRESS

opamp, capacitors, and low leakage switches. It integrates input small photoionization currents in the picoampere to nanoampere range in determined period, storing the resulting voltage on the integrating capacitor (section 2 of the Supplement 1). The signal is then sent from the transimpedance amplifier into the PC through the data acquisition system (C Series, NI−9401, 8 I/O USB−6001− NI). The software (LabVIEW2023) on this module controls the hardware module, collecting the data for display and post-acquisition treatment. LabVIEW2023 was coded for the purpose of operating the experimental apparatus and at the current configuration it runs the mass flow controls (MFC) (Aalborg Instruments, USA) on the air sampling system and the signal acquisition. For this proof-of-concept experiment we used an air sampling system (depicted in section 3, Supplement 1) whereby the saturated vapor pressure (SVP) of the target compound at room temperature and atmospheric pressure was diluted with nitrogen using the MFCs.

### 2.5. Chemical compounds

Ethanolamine (IUPAC name: 2-aminoethanol, purified by redistillation, ≥99.5 %) and eucalyptol also known as 1,8-cineole (IUPAC name: 1,3,3-Trimethyl-2-oxabicyclo[2.2.2]octane, ≥99 %) were acquired to Sigma-Aldrich and used without further purification. $N_2$ was used as a carrier gas from a home-built supply. The ionization potential of ethanolamine listed in the NIST webbook is 8.96 eV. To our best knowledge the ionization potential of eucalyptol has not been reported in literature.

### 2.6. Background signal acquisition

Before testing with a VOC, we characterized the background signal of a predefined selected range of pump wavelengths, according to chemical tested (ethanolamine or eucalyptol). These measurements serve as a baseline signal to be compared with the photoionization signal.

### 2.7. Photoionization signal acquisition procedure

The signal acquisition followed a four-step procedure. Firstly, while the laser is OFF a flow of $N_2$ is passed into the ionization chamber for 30 s. Secondly, we turn ON the laser at the desired wavelength for 60 s. Thirdly, we open the valve of the air sampling unit, so the desired mixture of VOC/$N_2$ is flown into the ionization chamber for 300 s. After this, the valve of the air sampling unit is closed, and the laser remains ON for an added 100 s. All the experiments were repeated three times, at room temperature, atmospheric pressure and applying a bias voltage $V_{\text{bias}}$ of 300 V to create a driving electric field for an enhanced collection of the generated current.

## 3. Results and discussion

### 3.1. Theoretical optimization of THG intensity generated by suspended $SiO_2$ nanomembranes

An investigation into the dependence of THG intensity on $SiO_2$ nanomembrane thickness was carried out via analytical expression for the third harmonic intensity ($I^{3\omega}$) and via a finite-difference time-domain (FDTD) numerical model. FDTD simulations can be applied to more complex structures than suspended nanomembranes, i.e., evaluate THG generation of structures beyond the capabilities of an analytical approach. Here, both methodologies were applied to planar nanomembranes.

#### 3.1.1. THG intensity simulations via analytical expression

We derived the intensity of the third harmonic ($I^{3\omega}$) by considering its electric field. According to Wang et al. [21], the electric field of the transmitted third harmonic wave ($E_{3t}^{3\omega}$) can be expressed

Optics EXPRESS

as

$$E_{3t}^{3\omega} = E_{2b}^{3\omega} \cdot A \cdot e^{i\vec{k}_{2t}^{3\omega}\cdot\vec{l}} \cdot e^{i\left(\vec{k}_{2b}^{3\omega}-\vec{k}_{2t}^{3\omega}\right)\cdot\vec{l}} \tag{1}$$

where $E_{2b}^{3\omega} = \frac{\pi\xi^3}{\Delta\epsilon}\left(t_{12}^{\omega}E^{\omega}\right)^3$ is the bound wave electric field, $\Delta\epsilon$ is the dispersion, $\vec{l}$ is the thickness of the material, and $\vec{k}_{2b}^{3\omega} = 3\omega\frac{n_2^{\omega}}{c}$ and $\vec{k}_{2t}^{3\omega} = 3\omega\frac{n_2^{3\omega}}{c}$ are the wave vectors from the bound and transmitted waves, respectively. The intensity of the third harmonic ($I^{3\omega}$) is proportional to the square of the electric field ($E_{3t}^{3\omega}$) of the transmitted wave:

$$I^{3\omega} \propto \left|E_3 t^3 \omega\right|^2 \propto \left|e^{i\psi^{\omega}} - e^{i\psi^{3\omega}}\right|^2 \tag{2}$$

or, as in Konishi et al. [22], we can write it relatively to the phase angle $\Delta\psi$):

$$I^{3\omega} \propto \left|1 - e^{i\Delta\psi}\right|^2 \tag{3}$$

where

$$\Delta\psi = \psi^{\omega} - \psi^{3\omega} = \frac{6\pi l}{\lambda_{\omega}}\left(n^{\omega} - n^{3\omega}\right) \tag{4}$$

with $\psi^{\omega}$ being the phase of the pump wavelength, $\psi^{3\omega}$ the phase of the THG, l the thickness, $n^{\omega}$ and $n^{3\omega}$ are the refractive indices of the fundamental and third harmonic frequencies, respectively. Our system generates pump wavelengths at 340 nm to 440 nm range, so $I^{3\omega}$ was calculated within this range. Figure 3 shows the data obtained from the analytical expression in Eq. (3).

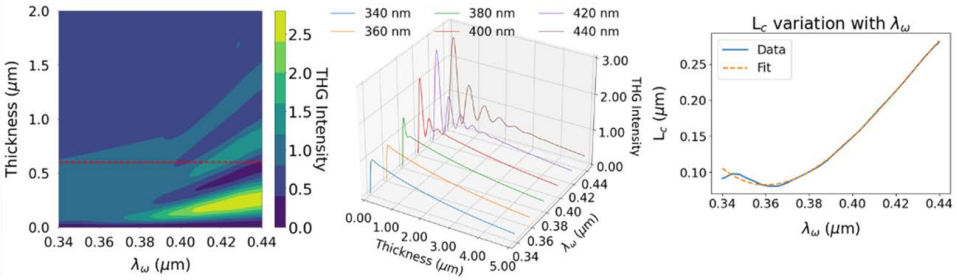



**Fig. 3.** Data obtained from Eqs. (3) and (4): (Left) Variation of intensity of the third harmonic ($I^{3\omega}$) with the thickness and $\lambda_{\omega}$; (Center) 2D overview of Eq. (3) for all simulated wavelengths; (Right) Variation of the coherent length ($L_c$) with the pump wavelength $\lambda_{\omega}$

From Fig. 3 (Left) and (Center), we can observe a sinusoidal behavior of the intensity relative to the thickness of the nanomembrane. The periodic nature of the intensity is related to the coherence length, $L_c$, which tells us the optimal thickness, for a pump wavelength, for which the intensity of the third harmonic is maximum since $\Delta\psi = \pi$ at this value.

Maximum intensity peaks can be found for thicknesses such as

$$L = 2mL_c \tag{5}$$

where m is a positive integer. The sinusoidal nature is "lost" and the intensity starts to decay linearly at longer thicknesses and can be explained by the longer path that the third harmonic wave travels, leading to higher scattering and diffraction losses. Moreover, we can see that the sinusoidal relationship starts to fade for smaller (more energetic) pump wavelengths. This can be explained by the increase in the extinction coefficient at smaller wavelengths, resulting in high absorption of the third harmonic wave associated with the bandgap of the material [21].

Figure 3 (Right) shows how the coherence length ($L_c$) changes with the pump wavelength ($\lambda_{\omega}$) for $SiO_2$, given by

$$L_c = \frac{\lambda_{\omega}}{6\left|n^{\omega} - n^{3\omega}\right|} \tag{6}$$

The dashed red line in Fig. 3 (Left) locates the THG intensity pattern for a thickness of 600 nm, used for our current application.

Optics EXPRESS

The intensity of the third harmonic is related to the phase angle expressed by Eq. (6), which itself depends on the refractive indices of the fundamental and third harmonic frequencies, $n^{\omega}$ and $n^{3\omega}$, respectively. These are generally complex quantities whereby

$$n^{\omega} = \eta(\omega) + i\kappa(\omega) \tag{7}$$

and

$$n^{3\omega} = \eta(3\omega) + i\kappa(3\omega) \tag{8}$$

where $\eta$ and $\kappa$ are the real and complex components of the refractive index related to the phase velocity and the optical extinction coefficient, respectively. The study by Konishi et al. [20] does not account for the optical extinction coefficient at the fundamental wavelength, as its values are negligible and can be disregarded. However, their analysis considered longer fundamental wavelengths compared to this work. The optical extinction coefficient for $SiO_2$ begins to increase at shorter wavelengths, which were not within the range explored by Konishi et al. [20] For that reason, where absorption starts to manifest itself, we considered the imaginary part of the fundamental wavelengths on our implementation, where is observed decay after the sinusoidal behavior.

### 3.1.2. THG simulations via FDTD (Lumerical)

Simulations via FDTD methods were applied using commercial tool Lumerical FDTD. The simulation set-up consists of the nanomembrane, a plane wave source at the pump wavelength corresponding to the laser, and a field time monitor to collect the electric field of the output signal (Fig. 4).

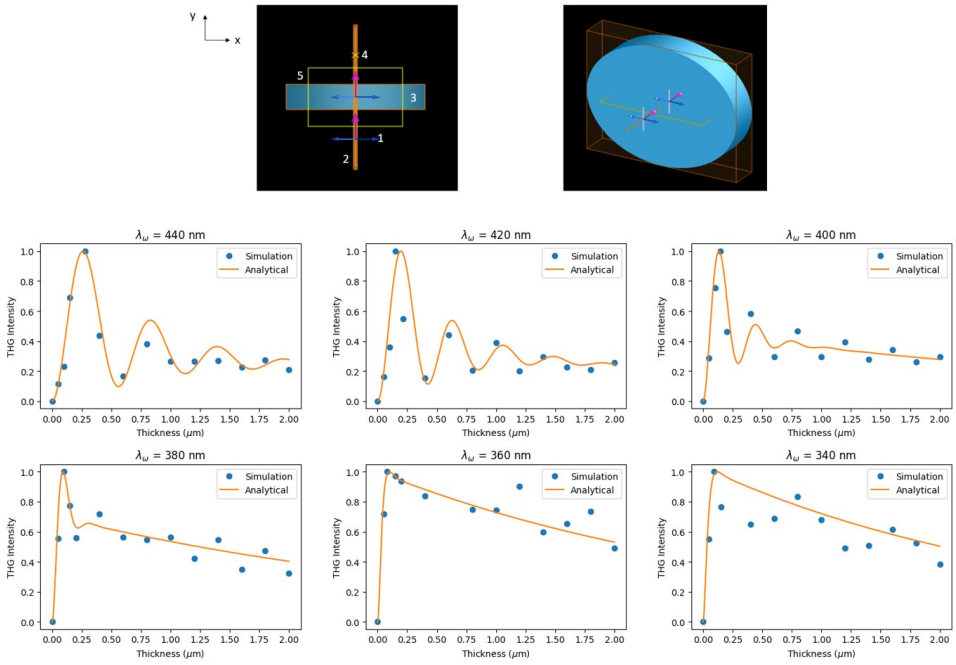



**Fig. 4.** Simulation setup: 1. Light source, 2. FDTD region, 3. Nanomembrane, 4. Field monitor, 5. Refractive index monitor. $I^{3\omega}$ vs thickness plots for the analytical expression (solid orange lines) and data retrieved from Lumerical (scattered points) for all simulated wavelengths.

Laser excitation was simulated through a quasi-monochromatic plane wave source. The laser electric field amplitude needed to be set as high ($\approx 10^9$ V/m) for the nonlinear effect to be

Optics EXPRESS

observed. The field time monitor collects the electric and magnetic fields of the output signal. Six different wavelengths between range340 nm to 440 nm were simulated; Figure 4 shows the relationship of the intensity of the third harmonic wave with the thickness of the nanomembrane for a given pump wavelength.

It is observed that the expected decaying sinusoidal relationship for all cases with the data retrieved from Lumerical ($I^{3\omega} \propto \left|E^{3\omega}\right|^2$ ) although the plots for $\lambda_\omega$ = 360 nm and $\lambda_\omega$ = 340 nm don't perfectly fit the data from the simulation. This can be attributed primarily to the $SiO_2$ refractive index reference data being retrieved from de Marcos et al. [23] and the fitting provided by Lumerical for the refractive index came down to a trade-off between the values for $\eta$ and $\kappa$, especially for the two lower wavelengths, since these values were very different from the raw data, hence the disparity for longer thicknesses. This was not a problem for longer wavelengths, as the data from the simulation revealed to be a good match with the analytical expression (3). Nevertheless, the relationship in Eqs. (3) and (5) is observed and thereby we can confirm that the FDTD simulation provides a result highly similar to the theoretical analytical evaluation of the same process.

### 3.2. Suspended $SiO_2$ nanomembranes characterization

The suspended $SiO_2$ nanomembranes over a silicon substrate created by nanofabrication processes suffer from compressive residual stress, resulting a buckled surface (Fig. S1b, Supplement 1). This stress is higher if the ratio of nanomembrane diameter to thickness becomes larger and hence represents a critical parameter limiting the lower bound of thickness achievable extended over a defined circular area. For the desired 600 nm nanomembrane thickness, a diameter of 1 mm was determined to be adequate for most membranes to maintain their structural integrity for the duration of our experimentation period. The optical transmittance and reflectance at visible wavelengths of fabricated 600 nm thick nanomembranes was evaluated for an array of nanomembranes and for individual nanomembranes (Fig. 4, Supplement 1). Results indicate that the nanomembranes exhibited a high transmittance. The optical properties observed for the entire configuration and the individual nanomembranes are consistent, particularly in the shape of the transmittance and reflectance curves and their absolute values. Slight variations in transmittance can be attributed to light being blocked by the bulk silicon that separates adjacent nanomembranes. These results demonstrate that optical characterization can be reliably performed on a single nanomembrane.

We extended their characterization to the verification of the formation of the THG signal. Due to experimental limitations, we report THG in the wavelength range of 390 nm to 460 nm using laser pump wavelength of 1190 nm to 1380 nm in steps of 10 nm. A simple imaging set-up (Fig. 5(a)) incorporating a CMOS camera (uEye, UI-1228LE-M) was positioned on the back of the detector chamber to verify whether the pump laser was aligned with the center of a $SiO_2$ suspended nanomembrane (Fig. 5(b)). For THG spectrally resolved measurements (Fig. 5(c)), we used a 10 x objective lens (Olympus) to focus the beam onto the sample and move it until it hits a $SiO_2$ suspended nanomembrane. The output light is then collected by a +40 mm focal length lens (Thorlabs, AC254-400-A) that goes to a dichroic filter acting as a separator of the fundamental beam that is transmitted to and absorbed by a beam blocker. The THG signal on the other hand is reflected into a fiber connected to the spectrometer (Ocean Optics, USB 2000+). While the pump laser has a reported spectral bandwidth of $\sim$ 10 nm, the spectral width of the THG signals observed is $\sim$ 3.7 nm across the wavelength range reported in Fig. 5(d). The spectra for pump wavelengths of 1280, 1290, 1360 and 1370 nm are not included due to interference of the THG with a spurious background signal present due to specificities in OPA operation. No significant signal fluctuations were observed in THG intensity, indicating the relatively stability of the THG process.

Optics EXPRESS

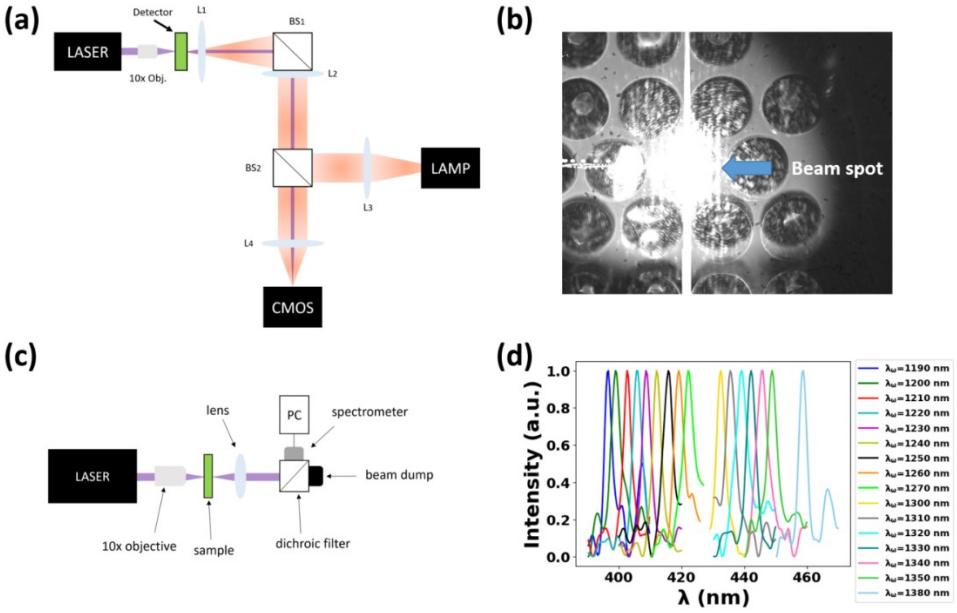



**Fig. 5.** THG produced by 600 nm thick $SiO_2$ nanomembranes in the range of 380 nm to 460 nm: (a) set-up for the alignment of the laser beam with respect to the nanomembrane device; (b) image of laser beam hitting the central part of a $SiO_2$ suspended nanomembrane; (c) optical setup for the detection of THG; (d) $SiO_2$ produced THG in the wavelength range of 390 nm to 460 nm using laser pump wavelength of 1190 nm to 1380 nm.

X-ray photoelectron spectroscopy (XPS) can be used to determine the bandgap of materials through the inelastic energy loss where a linear fit is applied to spectra near the approximate location of inelastic losses onset. The background level is then established through a Shirley background fitting. The onset energy of inelastic losses is identified by extrapolating the linear-fit line to its intersection with the background level [24]. The bandgap energy is calculated as the difference between the energy of O 1s core-level peak (532.4 eV) and the onset energy of inelastic losses (541.2 eV). Therefore, the bandgap energy is determined to 8.8 eV (section 7, Supplement 1).

### 3.3. *Signal acquisition*

We use ethanolamine and eucalyptol for the validation of our experimental apparatus. Ethanolamine is used in several industry processes, is the simplest and most abundant phospholipid headgroup and has gained interest as life precursor [25,26]. Eucalyptol is a monoterpene oxide [27] a component of essential oils derived from several plants species [28], present in everyday products [29–31] with unreported, to our best knowledge, first ionization potential. These compounds were chosen due to low safety risks and SVPs of 0.404 mmHg and 1.90 mmHg at 25 °C for ethanolamine and eucalyptol, respectively. Signal detection occurred at normal temperature and pressure (NTP). To attribute the current detected by our transimpedance amplifier circuit as VOC photoionization, we utilized the four step signal acquisition as detailed in Fig. 6(a). Applying this routine to ethanolamine and eucalyptol we acquired the signals shown in Figs. 6(d)-(g). Ethanolamine was excited with a wavelength at 390 nm while eucalyptol at 360 nm in order to attain photoionization.

Optics EXPRESS

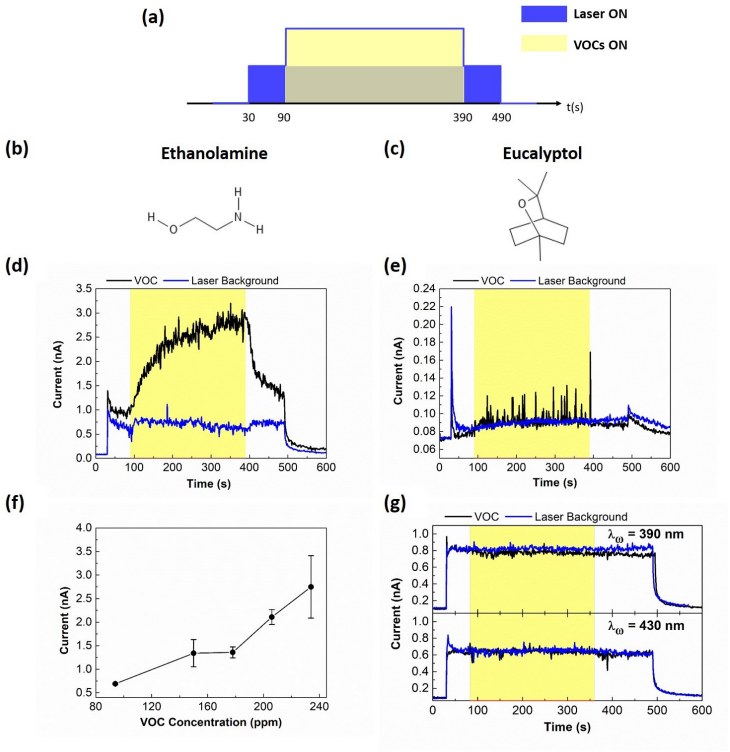



**Fig. 6.** (a) Schematic drawing for the acquisition sequence of the photoionization signal; (b) ethanolamine chemical structure; (c) eucalyptol chemical structure; (d) representative current variation for the laser background (in blue) and with ethanolamine (178 ppm) at the predefined times (in black) excited at 390 nm; (e) current variation for the laser background (in blue) and with eucalyptol (711 ppm) at the predefined times (in black) excited at 360 nm; (f) dependence of the photoionization signal with ethanolamine concentration ranging from range 94 ppm to 234 ppm; (g) current variation for eucalyptol (in black) for pump wavelengths at 390 nm and 430 nm.

### *3.4. Laser-induced current background signal*

Since we are dealing with volumetric gas fluxes, current responses are not guaranteed to be instantaneous. Hence, stabilization periods have been incorporated in the photoionization signal acquisition procedure. After switching on the laser illumination on our nanomembranes, we observe a current variation of 1.3 nA at 390 nm and 0.14 nA at 360 nm indicating the generation of a laser-induced current in the $SiO_2$ nanomembrane. Similarly to linear absorption in semiconductors, electronic excitations in dielectrics, which have a larger electronic bandgap, can also take place, provided that the relationship $h\nu \geq E_g$ holds ($h = 6.626 \times 10^{-34}$ JHz$^{-1}$ is Planck's constant, $\nu$ is the frequency of light, and $E_g$ is the bandgap of the material). However, if the incident electric field is very strong, such as in the case of ultrafast lasers, multiple photons can be absorbed by a material that is otherwise transparent at the corresponding wavelength through multiphoton absorption, and electronic excitations in the material take place, if $nh\nu \geq E_g$, where $n$ is the number of absorbed photons. The type of ionization in a strong-field regime can be classified as either multiphoton and tunneling, and can be determined by evaluating the Keldysh

Optics EXPRESS

parameter $\gamma$ given by [32]

$$\gamma = \frac{2\pi c}{\lambda}\frac{\sqrt{mE_g}}{|e|E} \tag{9}$$

where $c = 3\times10^8$ ms$^{-1}$ is the speed of light in vacuum, $\lambda$ is the incident wavelength, $m = 9.1\times10^{-31}$ kg and $|e| = 1.6\times10^{-19}$ C is the mass and the absolute charge of the electron respectively and $E$ is the incident electric field. We calculated the Keldysh parameter at $\lambda = 400$ nm, yielding a value of $\sim 46$, corroborating the presence of the multiphoton regime. We use the average output power instead of the electric field, as this is the value we can measure and monitor. Furthermore, we evaluated this hypothesis by carrying out three experiments (section 6, Supplement 1). In the first one, we studied the real-time response of the nanomembranes in respect to increasing input power levels. We observed a nonlinear increase in the current, which supports that a high density of high-power photons can induce a current in a dielectric material via multiphoton ionization. In the second experiment, the input power was held at its maximum power for the same wavelength, while $V_{\text{bias}}$ was varied from 0 V to 300 V in increments of 50 V. The results show an increase in the signal with higher applied $V_{\text{bias}}$, indicating improved current collection under strong electric fields. Therefore, by applying a lower $V_{\text{bias}}$ can reduce the laser background, albeit at cost of decreased current collection. Reducing the distance between the metal plates in the $SiO_2$ nanomembrane device represent a solution to maintain or even enhance the driving electric field while applying a lower $V_{\text{bias}}$. Finally, we found out a strong correlation between the output power of the laser at different wavelengths ranging from 320 nm to 460 nm and the induced current within the nanomembranes. Taking into account the bandgap of our nanomembranes (8.8 eV), reabsorption of VUV photons can also be occurring below 422 nm along the multiphoton absorption considered, resulting in the generation of electron-hole pairs [20].

### 3.5. THG efficiency induced by $SiO_2$ nanomembrane

To estimate THG efficiency, we analyse photocurrent measurements resulting from photoionization. Assuming a quantum efficiency of unity, each ionized molecule produces one electron per vacuum ultraviolet (VUV) photon. The experiment involved illuminating the nanomembrane at 360 nm light and monitoring the resulting current over 300 s. The incident light's average power, measured post-focusing lens, was 104 mW at a laser repetition rate of 60 kHz.

The energy per pulse is determined as:

$$E_{\text{pulse}} = \frac{P_{\text{avg}}}{\text{Rep Rate}} = 1.733 \quad \mu\text{J} \tag{10}$$

The number of incident photons per pulse is obtained by dividing this energy by the energy of a single photon at 360 nm:

$$N_{360nm} = \frac{E_{\text{pulse}}}{E_{360\text{nm}}} = 3.139\times10^{11} \quad \text{photons/pulse} \tag{11}$$

Over 300 s, at a repetition rate of 60 kHz, the total number of incident photons is $5.65\times10^{18}$. To estimate the number of VUV photons, we reference the photocurrent measured with eucalyptol, which stabilized at 22 pA. The corresponding total charge over 300 s is:

$$Q = I\Delta t = 6.6\times10^{-9} \quad \text{C} \tag{12}$$

Dividing this charge by the elementary charge $|e| = 1.6\times10^{-19}$ C yields the total number of electrons, which, under the assumption of unity quantum efficiency, corresponds to the total

Optics EXPRESS

number of VUV photons:

$$N_{120\text{nm}} = \frac{Q}{|e|} = 4.13 \times 10^{10} \quad \text{VUV photons} \tag{13}$$

The THG efficiency is then given by the ratio of generated VUV photons to incident fundamental photons:

$$\eta = \frac{N_{120\text{nm}}}{N_{360\text{nm}}} = 7.3 \times 10^{-9} \tag{14}$$

This efficiency is consistent with reported literature values [33]. However, it represents an upper bound, as the actual THG efficiency is likely lower due to assumption of and ideal quantum efficiency and other loss mechanisms.

### 3.6. VOC photoionization current

In the VOC gas exposure phase of the photoionization signal acquisition process (Fig. 6(a)), an increase in the current signal is observed after exposing the backside of the $SiO_2$ nanomembrane to a gas flow containing the SVP at normal pressure and temperature (NPT) of ethanolamine (Fig. 6(d)) and eucalyptol (Fig. 6(e)), or fractions thereof diluted in a $N_2$ flow. This signal increase is only observed when exposing the nanomembranes to a THG pump wavelength for VUV whereby the VUV photon energy is higher than the first IP of the chemical compound. For ethanolamine, with a first IP of 8.96 eV ≈ 138.7 nm, this wavelength was set to 390 nm, corresponding to THG at 130 nm, whereby a variation of the photocurrent is observed in Fig. 6(d,f). For a pump wavelength of 420 nm generating THG at 143 nm, *i.e.* below the IP of ethanolamine, no change was detected in the current signal (section 8, Supplement 1). A similar behavior was observed for eucalyptol, whereby a pump wavelength of 360 nm, corresponding to THG at 120 nm, generated a photocurrent, while pump wavelengths of 390 nm to 430 nm no current variation was observed. We consider that the threshold behavior whereby photocurrent is observed only at wavelengths more energetic than the first ionization potential, provides an indirect proof for the generation of VUV at 130 nm to 120 nm as a dominant mechanism leading to photocurrent detection of VOC ionization over other nonlinear effects, including SHG, higher harmonic generation (HHG, 4+), self-focusing and self-phase modulation. Namely, since our $SiO_2$ nanombembranes have been fabricated with PECVD, they are amorphous and can be considered noncentrosymmetric, potentially leading to the generation of SHG. However, SHG is not capable of generating photoionization current since the SHG photon energy is well below the ionization potential in the experiments with ethanolamine and eucalyptol reported in Fig. 6. Likewise, potential HHG processes also fail to induce any photocurrent (section 8, Supplement 1), most likely due to HHG conversion efficiencies being well below the one defined for THG above.

The THG conversion efficiency of the $SiO_2$ nanomembranes into VUV light is estimated as $7.3 \times 10^{-9}$. Nonetheless, our results indicate sufficient VUV is generated to induce the photoionization of ethanolamine and eucalyptol molecule, generating an electron and a cation. An electric field created between both electrodes separates the electron and ion, inducing a current amplified and converted into a voltage signal by the integrating transimpedance amplifier. With a current integration time of 200 ms and a signal acquisition rate of 1 Hz, the photoionization signal from ethanolamine is sustained, while the signal obtained from eucalyptol photoionization is observed in bursts. Given the higher SVP of eucalyptol with respect to ethanolamine the latter might be attributed to increased difficulty of our set-up to induce photoionization at 10.3 eV, due to decreased pump laser output and VUV photon reabsorption by $SiO_2$. As expected, when the VOCs are removed from the gas flow entering the ionization chamber, the current goes back to the laser background level, albeit slower than expected based on the gas volume renewal timeframe in the ionization chamber of less than 1 s, considering the volume of 2427.3 $mm^3$ and exchange gas flow rate of 200 mL $min^{-1}$. Finally, a near linear dependence on gas concentration was found for

Optics EXPRESS

ethanolamine as illustrated in Fig. 6(f), changing concentration levels from 94 ppm to 234 ppm through dilution of ethanolamine SVP in a $N_2$ flow, maintaining a constant total flow rate of 200 mL $min^{-1}$. Concentration values depicted take into consideration the laser background.

## 4. Conclusions

We have successfully integrated a pulsed (60 kHz, ≈230 fs), wavelength tunable (340 nm to 440 nm) laser with suspended $SiO_2$ nanomembranes of 600 nm thickness for efficient THG in the VUV. We demonstrated that this tunable quasi monochromatic VUV is capable of photoionizing molecules in the gas phase derived from SVP concentrations of ethanolamine and eucalyptol. Furthermore, through the photoionization of eucalyptol at 360 nm THG pump laser illumination we indirectly demonstrate the generation of VUV down to 120 nm, significantly lower than what has been demonstrated to date and with an associated energy of 10.3 eV above the 8.8 eV bandgap of $SiO_2$. The photoionization current was determined to be linearly dependent on the gas concentration for ethanolamine. Furthermore, our investigation places an upper and lower bound on the ionization potential of eucalyptol of 10.3 eV and 9.5 eV, respectively.

The detection efficiency of gases in our set-up relies critically on the efficiency of the THG process, determined to be on the order of $\sim 10^{-8}$. Optical electric anapoles or so-called dark modes are conditions attained in the illumination of nanostructures that strongly reduce the overall scattering through the destructive interference of the electric and toroidal dipoles. These anapoles have been reported to vastly increase the THG process utilizing illumination wavelengths in the near infrared [34]. Recently, we have reported through numerical simulations the creation of anapole modes with illumination wavelengths in the UV-A spectral region [35]. Hence, nanosized $TiO_2$ rectangular cuboids might provide a pathway in the future to drastically improve the gas sensitivity of our measurements set-up into the ppb detection range. Further future work will focus on the extension of the wavelength range down to 105 nm, the lower limit of our THG pump laser, exploring crystals such as $MgF_2$ and $CaF_2$.

**Funding.** HORIZON EUROPE Framework Programme (101057510); European Social Fund (ESF): Investing in Your Future (RYC2022-035710-I, MCIU/AEI/10.13039/501100011033); Xunta de Galicia (ED431F2024/15); Ministerio de Ciencia, Innovación y Universidades (MCIU) (BIPHOTO (CNS2023-145364)).

**Acknowledgments.** The authors thank the Micro and Nanofabrication Facility at INL for their support on the development of the suspended $SiO_2$ nanomembranes device, the Electron Microscopy and X−Rays Facility at INL, in particular, Oleksandr Bondarchuk for the XPS measurements and the Nanophotonics and Bioimaging Facility. Ana I. Gómez-Varela acknowledges the grant RYC2022-035710-I funded by MCIU/AEI/10.13039/501100011033 and by ESF Investing in your future and Xunta de Galicia (project reference ED431F2024/15). Martín López-Garcia thanks the MCIU of Spain through the grant BIPHOTO (CNS2023-145364). Disclaimer: Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

**Disclosures.** Rafael Vilarinho, Hedieh Mahmoodnia, Martín López-Garcia, Adelaide Miranda and Pieter A. A. De Beule have filed a patent application to Portuguese authority no. 20242006372381 covering this material and its use as a VOC sensor.

**Data availability.** The data that supports the findings of this study are available within the article and its Supplement 1.

**Supplemental document.** See Supplement 1 for supporting content.

## References

1. B. Edlén, "Wavelength measurements in the vacuum ultra-violet," Rep. Prog. Phys. **26**(1), 181–212 (1963).
2. C. Fröhler-Bachus, R. Friedl, S. Briefi, *et al.*, "Absolute radiometric calibration of a VUV spectrometer in the wavelength range 46–300 nm," J. Quant. Spectrosc. Radiat. Transf. **259**, 107427 (2021).
3. J. Berkowitz, E. Rühl, and H. Baumgärtel, "Valence Ionization Processes in the VUV Region," in *VUV and Soft X-Ray Photoionization*, U. Becker and D. A. Shirley, eds. (Springer US, 1996), pp. 221–261.
4. C.-Y. Ng, "Vacuum Ultraviolet Spectroscopy and Chemistry by Photoionization and Photoelectron Methods," Annu. Rev. Phys. Chem. **53**(1), 101–140 (2002).
5. R. Jiang, D. Mou, Y. Wu, *et al.*, "Tunable vacuum ultraviolet laser based spectrometer for angle resolved photoemission spectroscopy," Rev. Sci. Instrum. **85**(3), 033902 (2014).

Optics EXPRESS


6. P. A. A. De Beule and A. Miranda, "Atmospheric Photoionization Detector with Improved Photon Efficiency: A Proof of Concept for Application of a Nanolayer Thin-Film Electrode," Sensors **21**(22), 7738 (2021).
7. J. C. White, H. G. Craighead, R. E. Howard, *et al.*, "Submicron, vacuum ultraviolet contact lithography with an F2 excimer laser," Appl. Phys. Lett. **44**(1), 22–24 (1984).
8. M. Yamaguchi, K. Ikeda, M. Suzuki, *et al.*, "Cell Patterning Using a Template of Microstructured Organosilane Layer Fabricated by Vacuum Ultraviolet Light Lithography," Langmuir **27**(20), 12521–12532 (2011).
9. I. A. Nevyazhskaya, N. V. Shilina, V. B. Shilov, *et al.*, "Development and production of UV and VUV sources," J. Opt. Technol. **79**(8), 521–523 (2012).
10. U. Kogelschatz, "Excimer Lamps: History, Discharge Physics, and Industrial Applications," (2004), pp. 272–286.
11. D. Basting and G. Marowsky, *Excimer Laser Technology*, vol. 364 (Springer, 2005).
12. V. Ayvazyan, N. Baboi, I. Bohnet, *et al.*, "Generation of GW Radiation Pulses from a VUV Free-Electron Laser Operating in the Femtosecond Regime," Phys. Rev. Lett. **88**(10), 104802 (2002).
13. F. Grüner, S. Becker, U. Schramm, *et al.*, "Design considerations for table-top, laser-based VUV and X-ray free electron lasers," Appl. Phys. B **86**(3), 431–435 (2007).
14. C. Brahms and J. C. Travers, "HISOL: High-energy soliton dynamics enable ultrafast far-ultraviolet laser sources," APL Photonics **9**(5), 050901 (2024).
15. J. E. Lovelock, "A Photoionization Detector for Gases and Vapours," Nature **188**(4748), 401 (1960).
16. J. Deiglmayr, H. Herburger, H. Saßmannshausen, *et al.*, "Precision measurement of the ionization energy of Cs i," Phys. Rev. A **93**(1), 013424 (2016).
17. D. Z. Kandula, C. Gohle, T. J. Pinkert, *et al.*, "Extreme Ultraviolet Frequency Comb Metrology," Phys. Rev. Lett. **105**(6), 063001 (2010).
18. P. J. Linstrom and W. G. Mallard, "The NIST Chemistry WebBook: A Chemical Data Resource on the Internet," J. Chem. Eng. Data **46**(5), 1059–1063 (2001).
19. E. A. Sosnin and D. A. Sorokin, "Current Sources of Vacuum Ultraviolet Radiation: State and Prospects (A Review)," Plasma Phys. Rep. **50**(7), 880–893 (2024).
20. K. Konishi, D. Akai, Y. Mita, *et al.*, "Tunable Third Harmonic Vacuum Ultraviolet Coherent Light Generation Using Dielectric Nanomembranes," in *22nd International Conference on Ultrafast Phenomena* (Optica Publishing Group, 2020), paper Tu4B.11.
21. X. H. Wang, D. P. West, N. B. McKeown, *et al.*, "Determining the cubic susceptibility ?(3) of films or glasses by the Maker fringe method: a representative study of spin-coated films of copper phthalocyanine derivation," J. Opt. Soc. Am. B **15**(7), 1895–1903 (1998).
22. K. Konishi, D. Akai, Y. Mita, *et al.*, "Tunable third harmonic generation in the vacuum ultraviolet region using dielectric nanomembranes," APL Photonics **5**(6), 066103 (2020).
23. L. V. R.-d. Marcos, J. I. Larruquert, J. A. Méndez, *et al.*, "Self-consistent optical constants of $SiO_2$ and $Ta_2O_5$ films," Opt. Mater. Express **6**(11), 3622–3637 (2016).
24. M. T. Nichols, W. Li, D. Pei, *et al.*, "Measurement of bandgap energies in low-k organosilicates," J. Appl. Phys. **115**(9), 094105 (2014).
25. V. M. Rivilla, I. Jiménez-Serra, J. Martín-Pintado, *et al.*, "Discovery in space of ethanolamine, the simplest phospholipid head group," Proc. Natl. Acad. Sci. **118**(22), e2101314118 (2021).
26. R. Ramachandran, M. Sil, P. Gorai, *et al.*, "Experimental and Computational Study of Ethanolamine Ices under Astrochemical Conditions," The Astrophys. J. **975**(2), 181 (2024).
27. S. Aparicio, R. Alcalde, M. J. Dávila, *et al.*, "Properties of 1,8-Cineole: A Thermophysical and Theoretical Study," J. Phys. Chem. B **111**(12), 3167–3177 (2007).
28. Z.-M. Cai, J.-Q. Peng, Y. Chen, *et al.*, "1,8-Cineole: a review of source, biological activities, and application," J. Asian Nat. Prod. Res. **23**(10), 938–954 (2021).
29. J. Schuberth, "Volatile Organic Compounds Determined in Pharmaceutical Products by Full Evaporation Technique and Capillary Gas Chromatography/Ion-Trap Detection," Anal. Chem. **68**(8), 1317–1320 (1996).
30. D. L. Capone, K. Van Leeuwen, D. K. Taylor, *et al.*, "Evolution and Occurrence of 1,8-Cineole (Eucalyptol) in Australian Wine," J. Agric. Food Chem. **59**(3), 953–959 (2011).
31. S. Goto, H. Suzuki, T. Nakagawa, *et al.*, "The Effect of Eucalyptol on Nursing Home Residents," Sci. Rep. **10**(1), 3996 (2020).
32. P. Balling and J. Schou, "Femtosecond-laser ablation dynamics of dielectrics: basics and applications for thin films," Rep. Prog. Phys. **76**(3), 036502 (2013).
33. N. Boroumand, A. Thorpe, A. M. Parks, *et al.*, "Keldysh ionization theory of atoms: mathematical details," J. Phys. B: At., Mol. Opt. Phys. **55**(21), 213001 (2022).
34. G. Grinblat, Y. Li, M. P. Nielsen, *et al.*, "Enhanced Third Harmonic Generation in Single Germanium Nanodisks Excited at the Anapole Mode," Nano Lett. **16**(7), 4635–4640 (2016).
35. D. Poeta, R. Vilarinho, M. López-García, *et al.*, "Optical anapoles excited by UV-A illumination," Opt. Lett. **49**(24), 7198–7201 (2024).